



Jaya Arora

Bachelor of Technology
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Mechanical Engineering

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SELECTED HIGHLIGHTS

- Shipped **3 AI apps** (Streamlit + HF Spaces + GitHub) — **Signal** (crisis-aware news reranker), **Grounding Harness** (LLM hallucination eval, F1 = 0.811), **Traffic Analytics** (real-time CV, 96.2% accuracy); **400+ DSA** on LeetCode/CodeChef.
- Focused on **search reranking, RAG, and LLM evaluation** with sentence-transformers, DistilBERT, NLI cross-encoders, and FAISS.

EDUCATION

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Motilal Nehru National Institute of Technology, Allahabad	7.2 / 10.0 (Till 4th Sem)	2024-2028
Secondary (X)	St. Joseph's Sr. Sec. School, CBSE	94.6%	2021

PROJECTS

- Signal — Crisis-Aware Search Reranker** Aug 2026
Python, PyTorch, HuggingFace Transformers, DistilBERT, sentence-transformers, Gradio, Streamlit [GitHub](#) | [Live Demo](#)
 - Built a **dynamic-weight reranker** — MiniLM bi-encoder for relevance, **hybrid credibility** ($0.6 \times$ publisher-prior + $0.4 \times$ DistilBERT classifier), exponential freshness decay. Mode-aware weight-flipping (Standard 0.75/0.10/0.15 → Emergency 0.45/0.25/0.30) triggered by a keyword-based crisis detector.
 - Fine-tuned **DistilBERT** on LIAR2 fact-checking — **75.72% val accuracy** on 12,520 samples using AdamW + linear warmup/decay on Apple Silicon MPS.
 - Curated an **80-article eval corpus** (BBC/Guardian/NPR + LIAR2) and a **70-outlet publisher trust table**; the blend fixed a real LIAR2 domain gap where the pure classifier ranked NPR below BuzzFeed on the same query.
- Grounding Harness — Hallucination Evaluation for LLM Citations** Jul 2026
Python, PyTorch, sentence-transformers, NLI cross-encoder, FAISS, HuggingFace, arXiv API [GitHub](#)
 - Built an **evaluation harness** verifying whether each claim in an LLM answer is entailed by its cited source — turns “hallucination-free” into a single trackable number.
 - Designed a **5-type fabrication taxonomy**; curated 25 hand-labeled claims across 5 arXiv abstracts. v2 (sentence-level retrieval + NLI) hit **F1 = 0.811** and **100% fabrication recall (15/15)**.
 - Shipped as a **repeatable regression check** with per-claim outputs and failure-mode analysis; open-sourced.
- Real-Time Traffic Analytics Pipeline** May 2026
Python, PyTorch, YOLOv8, ByteTrack, OpenCV, Gradio, HuggingFace Spaces [GitHub](#) | [Live Demo](#)
 - Built a **YOLOv8-nano + ByteTrack pipeline** for real-time vehicle detection; persistent IDs across 2,208 frames with dual-line crossing counter.
 - Designed the **eval pipeline vs. a manually annotated 2,208-frame ground truth** — **96.2% accuracy (100% recall)**; documented tracking vs. counting failure modes.
 - Deployed live on **HuggingFace Spaces** (Gradio + matplotlib dashboard) with annotated MP4 + per-event CSV log.

TECHNICAL SKILLS

- AI/ML:** PyTorch, torchvision, HuggingFace Transformers, DistilBERT, MiniLM, YOLOv8 (Ultralytics), ByteTrack, OpenCV, Scikit-Learn, NumPy, Pandas
- LLM & Agent Stack:** RAG pipelines, sentence-transformers, NLI cross-encoders, FAISS, LLM APIs (Claude, OpenAI), evaluation harnesses
- Backend & Deployment:** FastAPI, Docker, Streamlit, Gradio, HuggingFace Spaces, Streamlit Community Cloud, Git LFS, Git, GitHub, matplotlib
- Languages:** Python, C++, C, TypeScript, JavaScript, HTML/CSS
- Interests:** LLM Evaluation, RAG, Search Reranking, Computer Vision, NLP, MLOps, DSA
- Competitive Programming:** Solved **400+ problems** across LeetCode and CodeChef
- Coursework:** Introduction to AI/ML, Optimization Methods in Engineering

POSITIONS OF RESPONSIBILITY

- Team Member — GPS & Route Planning**, Self-Driving Car Project, MNNIT Allahabad Jun 2026 - Present
- Coordinator**, Taekwondo Club, MNNIT Allahabad Jan 2026 - Present
- Networking Team Member**, E-Cell MNNIT Feb 2025 - Nov 2025
- Organising Team Member**, TEDxMNNIT 2025 Mar 2025 - May 2025

ACHIEVEMENTS

- 1st Prize — Marketing Mavericks**, GTM Strategy Competition, Renaissance 2025, E-Cell MNNIT 2025
 - Pitched a **GTM strategy** covering segmentation, pricing, and customer acquisition.
- Silver Medal — Taekwondo**, Spardha 2025, Annual Sports Fest, IIT (BHU) Varanasi 2025